

When a CT is used, the current transformer ratio (CTR = primary current/secondary current) must be specified correctly in the program. If the CT is not used, the software's CTR value should be set to 1.0.

If a current transformer is used, the CT secondary must never be kept open-circuited while the primary is energised, as this may damage the CT, the circuit board, and pose a safety hazard.

The control circuit is powered by 5V DC from the Arduino Nano, supplied via a 12V, 1A adaptor. The output of the ACS712 sensor module

is connected to the Arduino Nano's A0 analogue input.

Working principle

The Arduino Nano continuously samples the sensor output and computes the RMS value of the AC load current. This measured current is compared with the preset full-load current value (I_s). Based on the plug setting multiplier (PSM) and time setting multiplier (TSM), the controller calculates the corresponding operating time using inverse definite minimum time characteristics stored as lookup tables in the program.

When the measured current exceeds the set threshold, the Arduino activates an on-delay timer and subsequently energises an output relay after the calculated delay. This relay can be wired to either raise an alarm or trip the load or feeder, depending on the protection requirement. The relay operation strictly follows the inverse-time characteristics defined in the software.

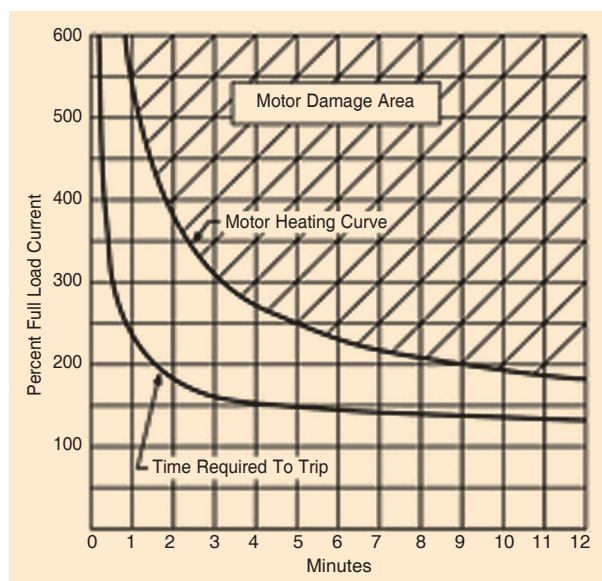


Fig. 3: Inverse-time characteristics (Source: Datasheet)

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IDMT_RELAY.ino
1 // Over current /IDMT(Inverse definite minimum time)relay based on Arduino Nano and Current sensor ACS712.
2 // Written by P.Balasubramanian. 24.5.2025
3
4
5 #include "ACS712.h"
6 ACS712 sensor( ACS712_30A, A0); //ACS712_05B for 5 Amp type, ACS712_20A for 20 Amp type, ACS712_30A for 30 Amp type
7
8 #include <elapsedMillis.h>
9 elapsedMillis timeElapsed;
10
11 int OLR = 12;
12 int resetbutton = 11;
13 float Is = 1.05 * 6.00; // current setting of relay x full load current of motor/feeder, in terms of CT primary current.
14 float psm; // Plug setting multiplier.
15 float tsm = 1.00; // Time setting multiplier, can be changed to get different inverse curves.
16 float lookupI[] = {1.00, 1.10, 1.20, 1.30, 1.40, 1.50, 1.60, 1.70, 1.80, 1.90, 2.00, 2.10, 2.20, 2.30, 2.40, 2.50, 2.60, 2.70, 2.80,
17 int lookupT[] = {254,117, 98, 69, 52, 45, 38, 34, 31, 28, 25, 25, 22, 19, 16, 13, 9, 6, 3, 3, 3, 3, 3, 3, 3, 3};
18 int x, m, s, resetbutton1;
19 float V, I, CTR = 6.00;
20
21
22 void setup()
23 {
24   pinMode(OLR, OUTPUT);
25   pinMode(resetbutton, INPUT_PULLUP);
26   Serial.begin(9600);
27   sensor.calibrate();
28 }
29
30 void loop()

```

Fig. 4: Snippet of the source code

Software

The firmware for the relay system is developed using Arduino IDE version 1.8.7. Before uploading the program, the following libraries must be installed in the IDE: the ACS712 library for accurate RMS AC current measurement, and the ElapsedMillis library for implementing the on-delay timing function.

The program file IDMT_RELAY.ino is uploaded to the Arduino Nano. In the code, the full-load current (I_s) is set to 6.0A, corresponding to the motor nameplate current, while the overcurrent pick-up level is defined as $1.05 \times I_s$.

Parameters such as I_s , CTR, lookupI, lookupT, and TSM may be modified to adapt the relay characteristics to specific application requirements. Fig. 4 shows a snippet of the source code.

Construction and testing

First, the source code (IDMT_RELAY.ino) is uploaded to the Arduino Nano. The circuit is then assembled on a small multi-purpose PCB and mounted inside a suitable enclosure, as shown in Fig. 5.

The current-carrying terminals of the ACS712 module must be properly insulated and connected to screw terminals. Adequate spacing must be maintained between low-

voltage control circuitry and high-voltage power wiring.

For testing, the ACS712 module's current input terminals are initially shorted to perform calibration. The circuit is powered on and the current reading is observed on the Arduino IDE serial monitor. The displayed current should be zero. After calibration, the shorting link is removed.

The system can be tested using a known 50Hz AC current source. The measured current should match the actual